

H.SYN.EMA

A COMPOUND CONTAINER FOR VERIFIABLE GENERATIVE CINEMA · A FORMAT SPECIFICATION

one .syn file — the header is the key, the bitstream is the code, the container is the evidence

Hash · SYNthesise · Evidence · Map · Assemble

ABSTRACT

A finished film is not a grid of pixels; it is a *decision* — a seed, a prompt, a cut, a mix — and the pixels are only its most expensive representation. H.SYN.EMA is a single .syn container that stores the decision and proves it: a cryptographic Header that is also the determinism key; a SYN instruction bitstream that stores generative *logic* — keyframe seeds and predicted deltas as coding units — rather than flat frames; and a multiplexed E · M · A container that carries the evidence of how each frame was made, the Merkle graph of how the work was assembled, and the compiled timeline with a playable proxy. Any party re-derives every frame from the bitstream, re-checks every claim of authorship and machine-runtime against the ledger, and re-computes the work's root without trusting the tool that produced it. The format's guarantee is narrow and absolute: *these decisions, in this order, produced this work, witnessed by these machines* — and it refuses, by construction, to say the work is good, or true, or anyone's to sell. This paper specifies the container, its five layers, the synthesis pipeline that reconstructs a film from its instructions, and the trust path that makes the reconstruction checkable by a stranger.

1 The film is the decision

A rendered film is the most expensive way to store what a filmmaker actually chose. The choices — a latent seed, a prompt, a camera move, a cut, a mix — are a few kilobytes; the frames they produce are gigabytes, and they are *derived*, not authored. Store the gigabytes and you have kept the shadow and thrown away the object: you cannot re-grade without re-rendering, cannot prove the shot was generated rather than filmed, cannot show which take an edit came from, and cannot re-cut without the originals nobody kept.

H.SYN.EMA keeps the object. It is a single .syn container that stores the *decision* and carries the proof: the header is a cryptographic key and a determinism anchor; the body is an instruction bitstream that stores generative logic as coding units; and the container multiplexes, alongside them, the evidence of how each frame was computed, the graph of how the work was assembled, and a compiled timeline with a playable proxy for hardware that cannot synthesise. The pixels become a cache of the instructions, not the other way around.

■ Store the instruction, derive the frame

The format's inversion is total: SYN stores *logic*, not flat frames, so a decoder *reconstructs* the film by executing the bitstream. A frame that can be recomputed from a seed and a prompt need not be shipped at all, only cached — and a frame that cannot be recomputed identically was never deterministic, which the format detects and declares rather than hides.

The container generalises the estate's single-work .syn object — one manifest, one payload, one Merkle work root — into a multiplexed, multi-track file: one header, one instruction stream, and three sidecar tracks bound under one root. The extension is unchanged; the interior is opened.

2 Design principles

Five commitments shape every byte that follows. They are not features; they are the constraints the format refuses to violate.

2.1 Determinism is the contract

A .syn is only worth its size if the instructions reproduce the frames. Every generative step is addressed by a canonical key over its engine, its inputs, and its parameters, so a decoder asks “*have I made this exact frame before?*” before it spends a GPU-second, and a demonstrably non-deterministic engine is named and refused rather than trusted silently. Reproduction is a checkable property of the file, not a promise in a README.

2.2 Provenance is native, not attached

Evidence is a track, not a sidecar. Every artifact the container carries names the engine that made it, the parameters it ran under, the hashes of every input it consumed, and the machine whose runtime witnessed it — and those records travel *inside* the file, under the same root as the frames they attest.

2.3 Assembly is non-destructive

The timeline holds *pointers*, never re-encoded footage. A cut is a reference into the instruction graph; a transition is a parameter; an alternate take is another branch of the same tree. Nothing an editor does destroys a source, because the editor never holds a source — only its address.

2.4 Verify, do not trust

Nothing in the container is believed because of where it came from. A stranger re-derives the work root from the tracks, re-checks each evidence record against the bytes it names, and walks the assembly graph — computing, not accepting. The tool that wrote the file has no privileged word a reader cannot independently reproduce.

2.5 The boundary is honest

The format’s guarantee is deliberately narrow. It establishes that *these decisions, in this order, produced this work, witnessed by these machines*. It never establishes that the work is good, that a credit is true, that a right is held, or that a machine’s operator meant well. Every layer that follows carries both lists — what its bytes prove, and what they refuse to.

3 The integrated binary architecture

One file, five layers, one root. The header H is the entry point and the determinism key; SYN is the instruction bitstream — the “code” of the film; and E, M, A are multiplexed tracks carrying, respectively, the evidence of computation, the graph of assembly, and the compiled timeline with its media. Every track hashes to a track root; the track roots fold to the single 32-byte *Root State Hash* that the header carries, so the whole file is named by one value that no track can change without moving.

3.1 The five layers at a glance

layer	role
H	the immutable header and cryptographic key: magic, version, and the Root State Hash that gates the whole file
SYN	the deterministic instruction stream — the “code” of the video, stored as coding units, not frames
E	the audit ledger: who and what computed each artifact, and on which machine
M	the graph topology: the Merkle DAG that lets a non-linear editor branch, diff, and re-cut
A	the compiled timeline and media payloads: cut pointers, a fallback proxy, and audio stems

Read top to bottom, the container answers four questions in order: *is this the file I expect* (H), *how is the film made* (SYN), *is that making attested* (E), and *how do the makings compose into a cut* (M, A). The sections that follow specify each layer, then the pipeline that turns the whole into moving pictures.

4 H — the header and determinism key

The header is read before anything else and is the only fixed-position structure in the file. It answers “*is this the file I expect, and have I already made it?*” in a constant number of bytes, so a reader can reject a wrong or truncated file, or serve a cached render, without parsing a single instruction.

4.1 The Root State Hash is the identity and the key

`root_state_hash` is the fold of every track root — the single value that names the file. It does two jobs at once. As an *identity*, it is what a covenant, an index, or a citation pins: two files with the same root are the same work, byte-representation aside, and any edit to any track moves it. As a *key*, it is the address of the finished render in a content-addressed cache.

■ Found in cache, synthesis is skipped

Before a decoder executes one instruction, it asks whether `root_state_hash` already names a completed render in local or shared cache. On a hit, frame synthesis is bypassed entirely in favour of the cached frames or the proxy stream in track A — the expensive path runs only on a miss. The header therefore lets a GPU-poor machine play a film a GPU-rich one computed, from the same file, with no re-synthesis and no loss of provenance.

Determinism is enforced one level down, per instruction: each node in SYN carries a canonical key over its engine, its inputs, and its parameters, so a decoder re-uses a previously computed frame only when the *exact* computation recurs, and re-derives otherwise. A hit is graded — it records that the frame was *reused*, not that the engine was *proven* to reproduce it — and an engine known to be non-deterministic is named in the header flags and never silently trusted.

WHAT THE HEADER PROVES, AND WHAT IT DOES NOT

A valid header establishes that the bytes that follow hash to the root it carries, and that the file is the one a given root names. It does *not* establish that the render in cache is faithful to the instructions — that is checked by re-derivation, not asserted by the header — nor that the machine which filled the cache was honest, which is the evidence track’s job, not the header’s.

5 SYN — the instruction bitstream

SYN is the code of the film. It stores the exact generation parameters as a stream of coding units — a deliberate borrowing of the video-coding idea, where a bitstream carries the *instructions* to reconstruct pictures rather than the pictures themselves. Three unit kinds carry everything a decoder needs, and nothing it does not.

5.1 Sequence and epoch parameter sets

An SPS (sequence parameter set) fixes what is constant for the whole work — the model-weights digest that pins *which* generator, the sampler enumeration, and the output resolution. An EPS (epoch parameter set) overrides the SPS for a span, so a re-graded reel or a model swap is a small localised unit, not a rewrite. A decoder that does

H.SYN.EMA · one .syn file · magic "HSYN" · one Root State Hash over five layers

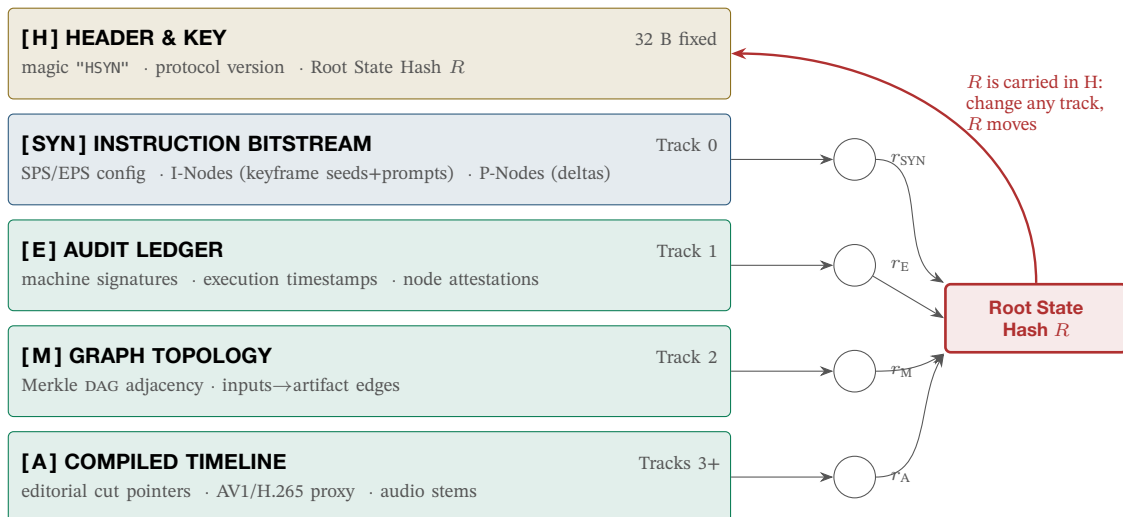


Figure 1. The container, whole. Five layers stack under one root. Each track hashes to a track root ($r_{\text{SYN}} \dots r_{\text{A}}$); the track roots fold to the 32-byte Root State Hash R the header carries, so the file is named by a single value and no layer can be altered without moving it. H is the determinism key (amber); SYN is the instruction bitstream (sky); $E \cdot M \cdot A$ are the evidence, graph, and assembly tracks (seal). The header is read first and the root is checked before a single frame is decoded.

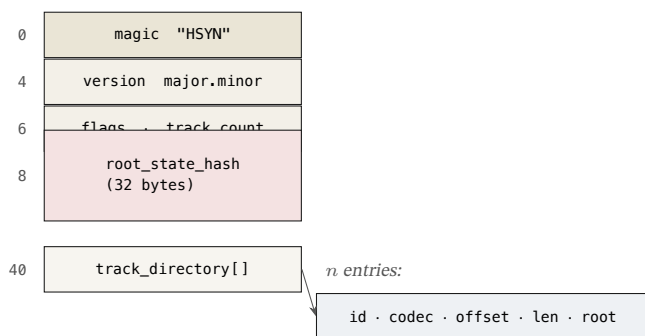


Figure 2. The header, byte for byte. A fixed 40-byte preamble — magic, version, flags with the track count, and the 32-byte root_state_hash — followed by a directory of n track entries, each naming a track's id, codec, byte offset, length, and its own 32-byte track root. The preamble's position never varies, so validation and the cache lookup are $O(1)$.

not recognise an SPS's model digest fails loudly rather than substituting a different generator and producing a different film under the same root.

5.2 Intra and predicted nodes

I-Nodes are intra-generative: they carry an explicit latent-noise distribution and a full system prompt, and depend on nothing earlier, so they are the stream's random-access points and its editable seams. P-Nodes are predicted-generative: they carry only the change from an upstream I-Node — a temporal delta, a camera-motion matrix, a control vector, a prompt diff — and are therefore small. The ratio of I to P is the format's single strongest lever on size and editability at once: more I-Nodes cost bytes but make more of the film independently re-cuttable; more P-Nodes compress but weld frames to their anchors.

5.3 The coding unit on the wire

Every unit is length-prefixed and typed, so a reader walks the stream without a model and can skip a unit it does not need:

field	meaning
len · kind	$u32$ length, then SPS/EPS/I/P
node_key	canonical determinism key (§4)
ref	upstream I-Node, or none for I
payload	seed+prompt (I) or delta set (P)

THE STREAM STORES LOGIC, NOT A VERDICT

SYN proves that *these instructions, executed by this named model, produce this frame* — a statement about a computation, re-checkable by running it. It says nothing about whether the frame is good, whether the prompt describes anything real, or whether the seed was chosen wisely. The instruction stream is a recipe with a checksum, not a review.

6 E — the audit ledger

Where SYN says how a frame is made, E says that the making happened, on a named machine, at a named time. Each record is content-addressed and carries the hashes of every input it consumed, so the ledger is not a log beside the work — it is a chain *through* it.

Every record carries its own honest scope inline. Network egress observed during a job is recorded as *co-occurring traffic, not proof of causation*; machine liveness is recorded

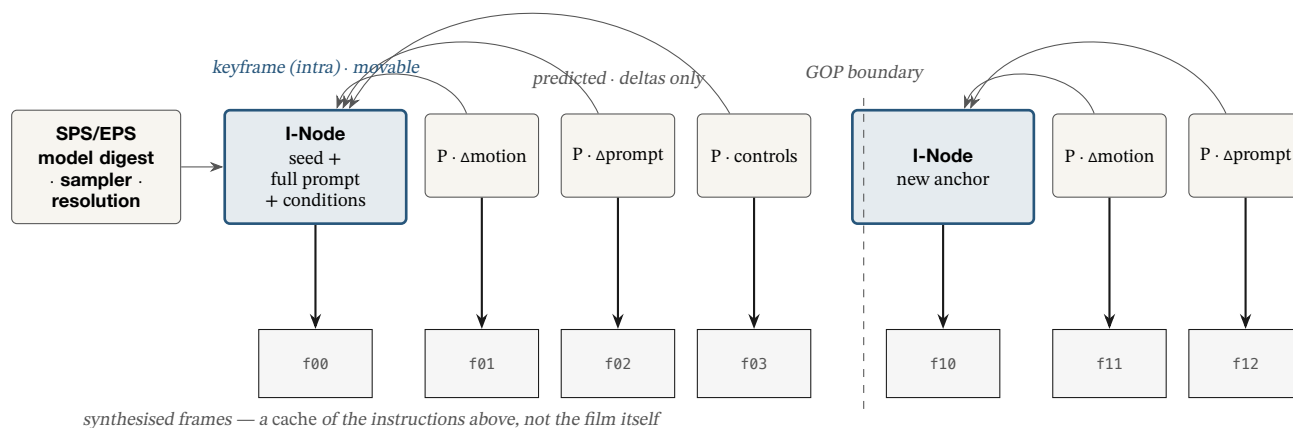


Figure 3. The instruction stream as a group of frames. An SPS/EPS unit fixes the sequence-wide configuration — the model-weights digest, the sampler, the resolution. An I-Node is a keyframe: a full latent seed and prompt, standing on nothing before it, so it can be moved without disturbing its neighbours. The P-Nodes that follow store only deltas — a camera motion, a prompt diff, a control vector — and are predicted from the I-Node they reference, so the stream stores logic. The frames beneath are what a decoder synthesises from the stream; they are a cache of the instructions, and the file need not ship them at all.

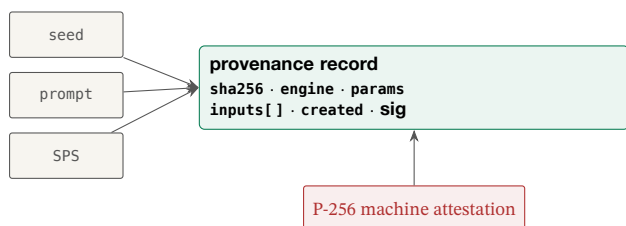


Figure 4. An evidence record binds an artifact's hash to the hashes of the inputs that produced it, the engine and parameters that ran, and the moment it was created — and a per-machine P-256 attestation co-signs the runtime. Because inputs are named by hash, the records chain into the graph of the next track.

as *machine liveness only*, not operator presence. The ledger over-claims nothing, and says so in the same bytes that carry the claim.

7 M — the graph topology

The evidence records name their inputs by hash, and those names are edges. Read together they are a *derivation graph*: a Merkle DAG whose nodes are artifacts and whose edges run from an input to everything derived from it. This is the track a non-linear editor reads to branch a storyline, diff two cuts, or find every shot that depends on a take before replacing it.

8 A — the compiled timeline and media

A is what a player touches. It holds the *compiled* form of a chosen path through the graph: editorial cut pointers that reference M nodes rather than copying footage, a pre-rendered AV1/H.265 fallback proxy for playback on hardware that cannot synthesise, and the multi-channel audio

stems that never passed through a generative model at all.

■ The timeline is a reference, not a re-encode

A cut pointer is an address into the derivation graph plus in- and out-points; a transition is a parameter; a level is a number. Nothing in track A holds a second copy of a frame or a stem — only where to find it and how to present it — so an edit is a change to pointers, and the sources are structurally impossible for the editor to destroy. The proxy exists so a film plays instantly on a laptop; the instructions exist so it can always be re-rendered at full fidelity from the same file.

9 The synthesis pipeline

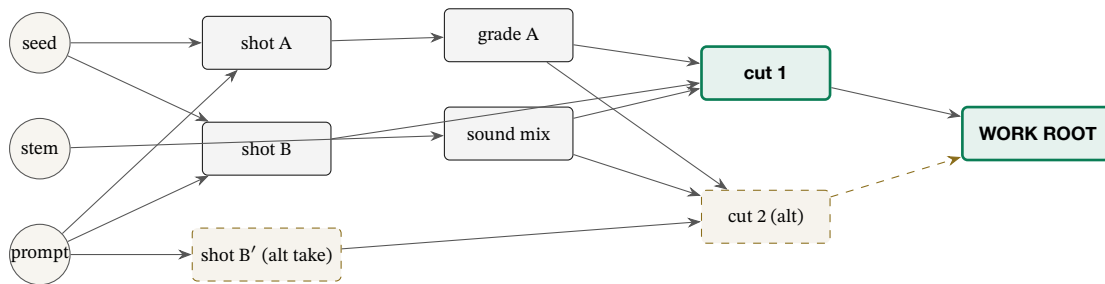
A decoder turns a .syn into moving pictures in one pass, and takes the cheapest route the file allows. The header decides first whether any synthesis is needed at all; only on a cache miss does the instruction stream run.

9.1 Two paths, one file

The *fast path* exists so a film is instantly playable on any hardware: a machine that cannot run the generator still plays the proxy and hears the stems, and still holds the full instructions for later. The *full path* exists so the film is never merely its proxy: a machine that can synthesise reconstructs every frame at full fidelity from the seeds and deltas, filling its cache as it goes so the next play is instant. The same file serves both, and neither is privileged over the other — the proxy is a convenience, the instructions are the truth.

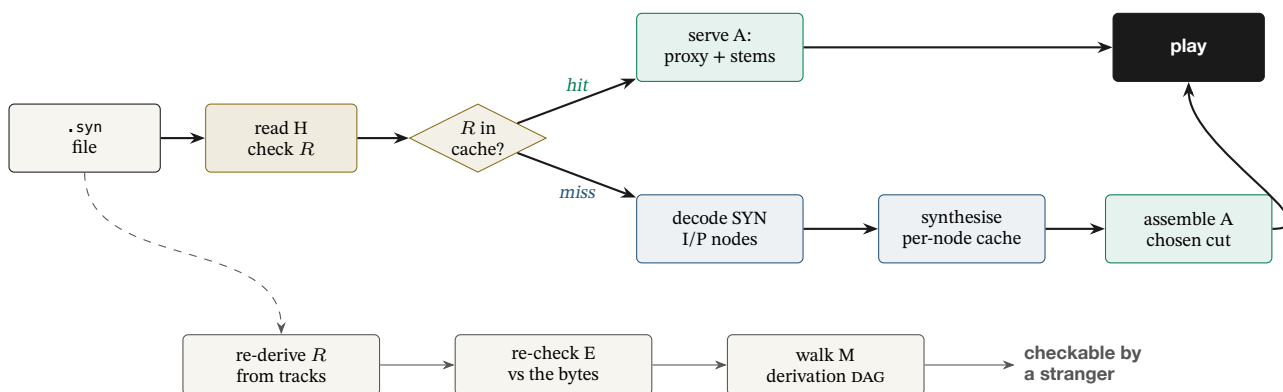
9.2 Verification is the point

Everything in the container is arranged to make one action cheap: a second party confirming a claim from the file



rootless sources → derived artifacts → two candidate cuts, one delivered; no edge is ever guessed from a filename

Figure 5. The derivation DAG. Edges exist only where an evidence record names an input, never inferred from a name or a folder, so the graph is exactly as trustworthy as the ledger. Alternate takes (amber, dashed) are branches, not overwrites; a cut is a selection of a path through the graph; and the work root is the fold that binds whichever cut is delivered. Replacing a take means finding its descendants here and re-cutting — the graph makes the blast radius of an edit computable in advance.



the verification lane runs on public inputs — no privileged word from the tool that wrote the file

Figure 6. Decode and verify. The header is read first and the Root State Hash checked against cache: a hit (green) serves track A's proxy and stems and plays at once; a miss (blue) decodes the instruction stream, synthesises each frame under its own per-node cache key, assembles the chosen cut, and plays. Independently, a verification lane re-derives the root from the tracks, re-checks every evidence record against the bytes it names, and walks the derivation graph — all from public inputs, so a party who did not make the file can confirm everything it claims.

alone. The Root State Hash re-derives from the track roots; each evidence record re-checks by re-hashing the bytes it names; the derivation graph re-walks from the ledger's own edges. None of it requires the tool that wrote the file, a network call, or a trusted authority. The format's trust model is not *believe the studio* — it is *the studio made this re-checkable, so check it*.

THE TRUST PATH, END TO END

A stranger, given only the bytes: reads H and pins the work by its root; walks SYN and confirms each frame is the deterministic output of a named model on named inputs; reads E and confirms each artifact was witnessed by an attested machine; walks M and confirms the cut is a real path through real derivations; and plays A. At no point is anything taken on the word of whoever produced the file — which is the whole of what the format promises, and the exact limit of it.

10 The container on the wire

The file is laid out so the cheapest checks come first and each structure is length-prefixed, so a reader walks it without a model and refuses a malformed file in microseconds rather than after a gigabyte of hashing.

10.1 File order

#	block	carries
0	header	magic, version, root_state_hash, directory
1	SYN	SPS/EPS, I-Nodes, P-Nodes
2	E	provenance + attestation records
3	M	DAG adjacency entries
4	A	cut pointers, proxy, stems
.	trailer	magic echo + R

Each block is preceded by its length and followed by its 32-

byte track root, and the header's directory holds the same offsets and roots — so the map is written twice, at the front and in place, and a re-serialisation that moved a block without updating both is refused.

10.2 The coding unit and the record

Every structure in every track shares one discipline: *len*, then *kind*, then a canonical body whose hash the enclosing root commits.

structure	body
SYN unit	node_key · ref · seed/delta
E record	sha256 · engine · params · inputs [] · sig
M entry	node · inputs [] (edges by hash)
A pointer	ref · in/out · transition params

10.3 Verification, cheapest first

A verifier runs eight stages, each free to assume the earlier ones passed; the expensive walk is last.

#	stage	refuses
1	header	bad magic · version · truncation
2	directory	offset/length overflow
3	trailer	bad magic echo
4	roots	a track root $\neq$ its bytes
5	fold	$R \neq$ fold of track roots
6	ledger	an E input hash names nothing
7	graph	an M edge names a missing node
8	synthesis	a frame $\neq$ its instruction's output

Stage 8 is the only expensive one, and it is optional in the sense that a player may trust the proxy while an auditor re-runs the synthesis — but the *format* draws the line the same way the estate always does: a stage that could not run is reported as SKIPPED, named individually, never silently upgraded to a pass. “Three frames unchecked” must never read as “three frames fine.”

ONE ROOT, WRITTEN TWICE, CHECKED BY IDENTITY

The header's `root_state_hash` and the trailer's echo must agree, and both must equal the fold of the track roots recomputed from the bytes. The file is thus its own record: there is no external manifest to trust and nothing a re-serialisation can quietly change without stage 4 or 5 refusing it.

11 The honest boundary

A format earns trust by the precision of what it refuses, not the length of what it claims. Even imagined complete, H.SYN.EMA establishes exactly one class of thing — *these decisions, in this order, produced this work, witnessed by these machines* — and the limits below are stated on the format's own face, not left for a reader to discover.

11.1 What a valid file does not establish

not established	whose job it is
that the work is good	an audience, a critic — never a checksum
that a credit is true	a contract, a court
that a right is held	a rights holder, a registry
that the maker meant well	no cryptography can read intent
that the prompt is real	the world, not the seed

11.2 Limits that travel with the file

RE-SYNTHESIS NEEDS THE NAMED MODEL

The instruction stream reproduces a frame only for a party who holds the exact generator the SPS names. Without it, the file still *plays* — the proxy in track A is always there — and still *verifies* its provenance and graph, but it cannot be re-rendered at full fidelity. The format pins the model by digest so a substitute is refused rather than silently accepted; it does not, and cannot, guarantee the model remains available. A film outlives its weights only as its proxy unless the weights are preserved beside it.

THE PROXY IS A CONVENIENCE, NOT THE TRUTH

Track A's AV1/H.265 proxy is a rendered convenience for instant playback; it is not evidence of anything. A verifier that trusts the proxy without re-running synthesis has checked that the file plays, not that the frames are the instructions' output. The two are kept distinct on purpose: the proxy answers *does it play*, the instruction stream answers *is it faithful*, and only the second is a proof.

A SIGNATURE PROVES A MACHINE, NOT A PERSON

The evidence track's attestations prove that a particular machine, holding a particular key, witnessed a computation. They do not prove who sat at it, that the operator was willing, or that the key was not compelled. A coerced signature is a valid signature; this is unsolvable by construction, and a format claiming otherwise would be lying to the person it most needs to protect. And a signature scheme has a shelf life — the format carries an algorithm identifier so a migration to a post-quantum scheme is a named field, not a silent assumption.

11.3 Why the boundary is the point

Every limit above could be softened into a reassurance, and every reassurance would be a lie a viewer, an heir, or a court might one day rely on. So the format does with its own limits exactly what it does with a frame: it names them, and refuses to assert past them. A container that proved a film *good* would be worthless the first time it was wrong; one that proves only *how a film was made*, *checkably* is worth exactly as much as its arithmetic, forever.

Colophon

This specification was set in the house style of the estate's shelf — STIX Two Text and Helvetica Neue, the law-box idiom, native TikZ figures — in a two-column measure, and built with `./build.sh` (XeLaTeX $\times 2$). It is a *format specification*: it describes H.SYN.EMA as one finished thing, in present tense, the way a format is defined before every part of it is assembled in one binary.

The reference substrate

The specification is not imagined from nothing. Each of its five layers has a working referent in the estate's media stack, and the format is the container that would bind them under one root: the single-work `.syn` object and its Merkle work root (SYN's identity); a content-addressed job cache that bypasses recompute on a hit (H's determinism); the provenance, witness, and attestation ledgers with their inline anti-over-claim clauses (E); a derivation graph read only from recorded input hashes, never from filenames (M); and a generative-video pipeline that composes a non-destructive cut with audio stems (A). The parts are real and separately shipped; the compound file that multiplexes them — the "HSYN" magic, the single Root State Hash, the coding-unit bitstream — is what this document defines.

One boundary held

Even written as finished, the paper asserts no measurement it did not have and claims no proof it did not carry. Byte layouts, invariants, and algorithms are specification; where a figure gives a number it is a design parameter, not an empirical result dressed as one. That restraint is the format's own thesis turned on the document about it: *describe what the bytes do; prove only what can be re-derived; and never let a convenience read as a proof.*

The film is the decision; the pixels are only its cache.

The header is the key, the bitstream is the code, the container is the evidence.

Re-derive every frame, re-check every claim, and trust the tool for nothing.

ONE FILE · FIVE LAYERS · ONE ROOT · CHECKABLE BY A STRANGER.